First-Jet Conditioning, Exact Coefficient Correction, and Power-Law Nodal Witnesses

A Scoped Ram Calculus Analysis of M675

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M675

October 8, 2026

Abstract

We analyze the finite-frequency mechanism in the manuscript *Power-law violations of Yau's nodal upper bound*. Its claimed conclusion concerns a single smooth metric and an infinite sequence of exact eigenfunctions; neither positive finite covariance nor a small differential residual establishes that conclusion. We compute the real five-dimensional first-jet Gram matrix of the source's four complex packets, retaining a smallest singular value of order $k^{-3/4}$ before envelope normalization. We prove a dimension-sensitive Gaussian mesh estimate and, using the source's packet and collar hypotheses, replace its noncancellation exponent 500 by 398, using a selected five-column volume. A homogeneous vector-quotient calculation then gives exact coefficient correction with residual order $M \geq 841 + 443R$, in place of the source's sufficient $2131 + 1088R$ for a prescribed finite C^R accuracy. A weighted resolvent criterion separates exact spectral persistence from finite-jet solvability. We rederive the sign-certificate and warped-product transfer and identify the uniform quantifiers required by the fixed-metric diagonal limit. These are restricted deductions conditional on the specified packet, envelope, and elliptic inputs, not a new independent proof of the full counterexample or a claim of historical priority. Contribution assessment: C3/5, explicit source-specific conditioning and correction improvements with proved local scope and an inherited global construction.

1 Problem, domains, and logical status

Let $B = S^4$ with round metric g_0 and density μ. A coefficient pair consists of a smooth symmetric positive contravariant tensor A and a smooth density factor $\rho > 0$. Put

$$L_{A,\rho} f = \rho^{-1} \operatorname{div}_\mu (A \mathrm{d}f), \qquad G^{-1} = A/\rho. \tag{1}$$

The closed nonnegative operator $-L_{A,\rho}$ acts on $L^2(B, \rho\mu)$; its quadratic form is $\int_B A(\mathrm{d}f, \mathrm{d}f)\,\mu$ on $H^1(B)$. Coefficient positivity and compactness give uniform ellipticity for each fixed pair. A sufficiently small common C^1 neighborhood of the round pair supplies the uniform geometric comparisons used below. Higher derivative constants need not be uniform over that neighborhood.

The source [1] asserts a fixed smooth metric on $S^4 \times S^1$ and real nonzero eigenfunctions with eigenvalues $\lambda_i \to \infty$ such that, for one fixed $\varepsilon_0 > 0$,

$$\frac{\mathcal{H}^4(\{u_i = 0\})}{\lambda_i^{1/2+\varepsilon_0}} \longrightarrow \infty. \tag{2}$$

We use the spectral parameter convention $-\Delta u = \lambda u$; the packet frequency is $k = \sqrt{N(N+3)}$, not λ. All assertions concerning one frequency mean sufficiently large k after fixing the input coefficients, collars, finite derivative orders, and requested tolerances. An unspecified C may depend on those data but not on k. Powers explicitly called uniform are independent of these finite choices.

Three levels of assertion. The source's envelope and packet construction and its global existence statement are inherited inputs, with their actual proof mechanisms examined in Sections 2 and 7. The Gram, mesh, quotient, and resolvent statements below are proved here under explicit hypotheses. The bounded computations accompanying the report check algebra and selected finite cases only. They do not establish any continuum lower bound, universal Gaussian event, spectral gap, or infinite smooth limit.

Private proof routes. The relevant Ram sources are the covariance and remainder arguments in core pattern 16 [2], the differentiated finite-jet and reconstruction proofs in the nonlinear integral-equation report [3], and the physical-metric compression discussion [4]. They enter as mathematical tools with their original restrictions, not as new names for the source theorem. The full public manuscript was read, including the exact correction and persistence proofs. The accompanying local read ledger records literal private line ranges and SHA-256 identities; it does not claim to have read the entire private corpus.

2 The source construction and its finite-order contracts

This section records the source-specific inputs needed for the improvements. It is not an independent replacement for every envelope induction.

2.1 The one-step quantifier and sign certificates

In round coordinates write $t = |y|^2$ and $u_* = (1-t)^{N/2}\cos(N\theta) = \Re(x_1 + ix_2)^N$. This is an exact round eigenfunction with eigenvalue $k^2 = N(N+3)$. For fixed $1/4 \le l_- < l_+ \le 1/2$, the source one-step proposition takes a smooth input pair in its common neighborhood, round on $t < l_+$, and arbitrary finite C^R accuracy. For every sufficiently large integer N it produces a nearby pair round on $t < l_-$, an exact simple eigenfunction at k^2, and a sign certificate of size at least $ck^{1+\gamma}$. The exponent $\gamma > 0$ is common; c and the frequency threshold may depend on the input and finite accuracy. This order of quantifiers is essential in the eventual diagonal argument.

A sign certificate is a finite collection of disjoint coordinate cylinders, each with two parallel closed three-dimensional end disks of opposite strict sign. Its size is the sum of their Euclidean three-dimensional disk measures. Strict signs on compact disks have a positive margin. Thus a sufficiently small C^0 perturbation and multiplication by any positive function preserve the certificate. Multiplication by a nonzero constant preserves its size, reversing the two signs if the constant is negative.

2.2 Envelope curvature and growth

Write $p = \nabla_G h$, $Q = (1 + |p|_G^2)^{1/2}$, and, for a unit vector $e \perp \mathrm{d}h$, set

$$J_h(e) = \nabla_G^2 h(p/Q, p/Q) + \nabla_G^2 h(e, e).$$

The source starts with a linear function of t and a local perturbation above $t = 1/2$ producing a favorable region. On prepared cubes of scale S, its gradient is nonzero, its Hessian is bounded by

$C|\nabla h|/S$, and a two-plane inside $\ker \mathrm{d}h$ has $J_h \geq c|\nabla h|/S$ in every unit direction. The two-plane requirement is stronger than positivity in one chosen direction.

The corrugation uses transverse radial wells ψ with radial derivative f, tangential curvature $u_0 = f/r$, and $f^2|f'| \leq Cu_0$, $f^2 \leq Cu_0$. In a parent cube the increment is $w = s\ell a\chi\psi(z'/\ell)$, with $s = |\nabla h|$ and $\ell = \nu^2 S$. On the plane defined by $\mathrm{d}(h+w)(e) = 0$ and $\sigma \cdot e' = 0$, the leading new curvature is $ca\chi u_0$, with error at most $\epsilon(\nu)(\chi u_0 + \nu^2)$. When the new curvature dominates, it supplies positivity. In the complementary regime the gradients are close and the old two-plane, after a small rotation, intersects the new two-plane in at least a line inside a three-dimensional hyperplane. The dimension calculation is $2 + 2 - 3 = 1$. The proof also treats the simultaneous vanishing of cutoff and well coefficients; a uniform positive lower bound on χu_0 is not available.

The mean slope gain in a well is $3\int_0^1 v^2\sqrt{1 + a^2F(v)^2}\,\mathrm{d}v > 1$. Packing and core selection are chosen so that the selected children gain relative to the *entire* parent mass. The derivative bound per generation is $\|D^j w\| \leq C_j s\ell^{1-j}$; disjoint supports do not contribute a factor equal to the number of cubes. With scale ratio $r = \nu^2$, slopes bounded by $C\beta^i$, and $m = \lfloor \theta \log k \rfloor$, the source obtains

$$\|h\|_{C^j} \leq C_j k^{\delta(j+1)}, \quad J_h \geq ck^{-\delta}, \quad \int_V Q\,\mathrm{d}x \geq ck^{\gamma}, \qquad \delta = 10^{-6}. \tag{3}$$

Here $\gamma = \theta \log(1+\zeta) > 0$ is fixed after the geometric construction. These are finite-frequency smooth envelopes, not an assertion that the envelopes converge in C^∞ as $k \to \infty$.

2.3 Packet jets and full remainders

At a normal-coordinate center choose $v \perp p$, $|v| = 1$. The complex vector $P = p + iQv$ obeys the *bilinear* identity $P \cdot P = -1$. The source solves for a complex symmetric Hessian S with $SP = 0$, $\nabla^2 h - \Re S \geq k^{-3\delta}\,\mathrm{Id}$, and $|S| \leq k^{20\delta}$. For $b = |p|/Q \geq k^{-12\delta}$ it imposes $b^2(\Re S)_{11} + (\Re S)_{vv} = 0$ and solves the compatible imaginary columns. For small b it uses a separate large negative transverse Hessian of size $k^{14\delta}$ and a Schur estimate. Dividing by b in this second regime, including $p = 0$, would be invalid.

For a homogeneous polynomial equation $q \cdot \partial H = F_m$, some coordinate q_i is bounded below because the *Hermitian* norm of q is bounded below. Integrating in that coordinate and using a finite nilpotent geometric series for the other derivative terms gives a right inverse on homogeneous polynomials. This does not require $q \cdot q$ to be nonzero. It supplies each successive eikonal coefficient and then each complete transport polynomial.

With $r_0 = k^{-100\delta}$, $X = \xi/r_0$, and $\kappa = kr_0 Q$, packets have form

$$w = e^{kh(z)} e^{\kappa\Phi(X)} \Big(\sum_{j=0}^{s} \kappa^{-j} a_j(X) \Big) \omega(k^{1/3} X).$$

The eikonal and transport polynomials are chosen to finite order T before letting k grow. On the cutoff annulus, Gaussian decay beats every fixed power. For every prescribed finite J, M one can choose T, s so that

$$|D^j w| \leq C_j k^{2j} e^{kh}, \qquad |D^j(\mathrm{div}_\mu(A\mathrm{d}w) + k^2\rho w)| \leq C_{J,M} k^{-M} e^{kh} \quad (0 \leq j \leq J). \tag{4}$$

One sufficient choice, read directly from the source remainder, is

$$T > 3(M + 200\delta + 2J + 3), \qquad s > \frac{M + 200\delta + 2J + 1}{1 - 100\delta}. \tag{5}$$

The source remainder before choosing these orders is bounded by $Ck^{200\delta+2J}(k^{2-T/3} + k^{1-T/3} + k^{-(1-100\delta)s})e^{kh}$. All differentiated remainders through J matter. A value-only eikonal error does not justify subsequent coefficient derivatives.

3 An explicit real first-jet Gram matrix

The relevant jet is real and has five rows, not a complex scalar variance. Set $\epsilon = k^{-3/4}$, $\eta = k^{-9\delta}$, and choose an orthonormal frame $e_1, \dots, e_4$, with $p = |p|e_1$ when $p \neq 0$. When $p = 0$, choose e_1 perpendicular to a favorable direction e_2. Define

$$v_2 = e_2, \quad v_3 = (e_2 + \eta e_3)/\sqrt{1+\eta^2}, \quad v_4 = (e_2 + \eta e_4)/\sqrt{1+\eta^2}, \quad p' = p + Q\epsilon e_1.$$

These are precisely the three transverse perturbations and one real slope perturbation used in the source. Write $b = |p|/Q \in [0, 1)$ and $r = \sqrt{1 + |p'|^2}/Q \in [1, 1+\epsilon]$.

Proposition 3.1 (Exact bank conditioning)**.** *Let C be the real 5×8 matrix of the real and imaginary parts of the four ideal normalized jets*

$$(1, be_1 + iv_2),\ (1, be_1 + iv_3),\ (1, be_1 + iv_4),\ (1, (b+\epsilon)e_1 + irv_2).$$

For $0 < \epsilon, \eta \leq 1$ and $0 \leq b \leq 1$, $r \geq 1$,

$$\sigma_{\min}(C) \geq \min\{\sqrt{3/11}\,\epsilon, \sqrt{2/7}\,\eta\}. \tag{6}$$

Here $\sigma_{\min}$ is the smallest of the five row singular values. For any real perturbation E of this matrix, $\sigma_{\min}(C+E) \geq \sigma_{\min}(C) - \|E\|$.

Proof. The Gram matrix CC^T splits between the value/e_1 coordinates and the e_2, e_3, e_4 coordinates. The first block is

$$B_R = \begin{pmatrix} 4 & 4b+\epsilon \\ 4b+\epsilon & 4b^2 + 2b\epsilon + \epsilon^2 \end{pmatrix}, \qquad \det B_R = 3\epsilon^2, \quad \operatorname{tr} B_R \leq 11.$$

Thus $\lambda_{\min}(B_R) \geq \det B_R / \operatorname{tr} B_R \geq 3\epsilon^2/11$. Writing $d = 1 + \eta^2$, the second block is

$$B_I = \begin{pmatrix} 1 + r^2 + 2/d & \eta/d & \eta/d \\ \eta/d & \eta^2/d & 0 \\ \eta/d & 0 & \eta^2/d \end{pmatrix}.$$

Its antisymmetric transverse eigenvalue is η^2/d. On the remaining two-plane the determinant is $(1+r^2)\eta^2/d$ and the trace is $1 + r^2 + (2+\eta^2)/d$. Consequently the smaller eigenvalue is at least

$$\frac{(1+r^2)\eta^2}{(1+r^2)(1+\eta^2) + 2 + \eta^2} \geq \frac{2\eta^2}{4 + 3\eta^2} \geq \frac{2\eta^2}{7}.$$

The antisymmetric eigenvalue obeys the same lower bound. For a unit row vector z, $\left\|(C+E)^T z\right\| \geq \left\|C^T z\right\| - \|E\|$; taking the infimum proves the perturbation claim without assuming a square inverse. □

Corollary 3.2 (The source bank near a net center)**.** *For the source's four packets the real normalized first-jet bank at a center has least row singular value at least $ck^{-3/4}$, and retains this bound within a sufficiently small fixed multiple of k^{-10}. The constants can depend on the fixed packet orders. In particular the source's displayed ck^{-1} bound holds there.*

Proof. The source center jets, normalized as $e^{-kh(z)}(w, \partial_\xi w/(kQ_z))$, differ from the ideal bank by at most $Ck^{-1+100\delta}$. Since $9\delta < 3/4$ we have $\eta \geq \epsilon$, and $k^{-1+100\delta} = o(k^{-3/4})$. The source spatial derivative bound on this normalized bank is Ck^3. Moving by Ck^{-10} therefore costs $O(k^{-7})$, again $o(k^{-3/4})$. Also $k|h(x) - h(z)| = o(1)$ on this distance, and fixed chart changes have uniformly bounded condition numbers. Passing to ordinary derivatives multiplies derivative rows by $kQ_z \geq 1$, which does not decrease the least singular value. The argument covers $b = 0$ without division by b. □

Proposition 3.3 (Selected-column volume)**.** *Select the real columns of packets one and four and the imaginary columns of packets one, two, and three. Their ideal* 5×5 *matrix* S *obeys*

$$|\det S| = \frac{\epsilon\eta^2}{1+\eta^2}, \qquad \left\|S^{-1}\right\| \leq C\epsilon^{-1} \quad (\eta \geq \epsilon). \tag{7}$$

For the actual source packets, at a center and within distance Ck^{-10}*, the ordinary, unnormalized selected first-jet matrix* S_x *therefore satisfies*

$$|\det S_x| \geq ce^{5kh(x)}k^{13/4-18\delta}. \tag{8}$$

Proof. In the row order value, e_1, e_2, e_3, e_4, the selected ideal matrix is block diagonal. The real block has columns $(1, b)$ and $(1, b+\epsilon)$, hence determinant ϵ and inverse norm at most C/ϵ for $0 \leq b \leq 1$. The imaginary block has columns v_2, v_3, v_4, determinant $\eta^2/(1+\eta^2)$, and inverse norm at most C/η, as seen by solving its triangular system. This proves (7). If E is the selected transport and spatial error, then

$$\left\|S^{-1}E\right\| \leq Ck^{3/4}(k^{-1+100\delta} + k^{-7}) = o(1).$$

Thus $|\det(S+E)| = |\det S|\,|\det(\mathrm{Id} + S^{-1}E)| \geq c|\det S|$. Restoring the value scale $e^{kh(z)}$ in all five rows and the derivative scale kQ_z in four rows multiplies the determinant by $e^{5kh(z)}(kQ_z)^4$. Here $Q_z \geq 1$, $e^{kh(z)} \asymp e^{kh(x)}$ nearby, and changes from the orthonormal center frame to any of the fixed charts have determinants bounded above and below. Since $\epsilon = k^{-3/4}$ and $\eta^2 = k^{-18\delta}$, (8) follows. No perturbed fourth imaginary column is used in this minor. □

3.1 Ram covariance and rank-loss interpretation

For independent standard real Gaussian coefficients g, a real jet $Cg + m$ has covariance CC^T; the deterministic beam only changes m. Adding independent families adds positive semidefinite matrices. For correlated coefficients one instead needs a bound for $C\Sigma^{1/2}$; the previous estimate follows from $\Sigma \geq s^2\,\mathrm{Id}$ only after multiplication by s. Positive semidefiniteness alone does not give a positive least eigenvalue.

The covariance proof in [2] is exactly $v^T \operatorname{Cov}(X)v = \mathbb{E}|v \cdot (X - \mathbb{E}X)|^2$. Its remainder estimate can be sharpened in the direction needed here. If $X = Y + R$ are defined on the same probability space and $(\mathbb{E}|R|^2)^{1/2} \leq e$, then

$$\lambda_{\min}(\operatorname{Cov} X)^{1/2} \geq \left(\lambda_{\min}(\operatorname{Cov} Y)^{1/2} - e\right)_+. \tag{9}$$

Indeed centering is an orthogonal projection in L^2, so for every unit v the reverse triangle inequality gives $\|v \cdot (X - \mathbb{E}X)\|_{L^2} \geq \|v \cdot (Y - \mathbb{E}Y)\|_{L^2} - e$. Taking the infimum proves the assertion. The same-law assumption is indispensable. Moreover a nonlinear remainder can destroy Gaussianity, so (9) alone does not preserve a Gaussian density bound. Our packet perturbation is a deterministic linear bank perturbation, and the jet remains Gaussian.

The private rank-loss example is also relevant: an almost surely nonzero uniform scalar on $[-1,1]$ has an inverse with infinite second moment. An almost sure full-rank assertion is not a quantitative inverse estimate. Here (6) and the continuum mesh estimate below supply actual margins. They are not inferred from sampled singular values.

4 Noncancellation with constants retained

Lemma 4.1 (A dimension-sensitive Gaussian mesh bound)**.** *Let a compact d-dimensional manifold be covered by finitely many fixed charts. Let J_k be a continuous $\mathbb{R}^m$-valued random field, $m > d$. On an event B_k assume $\|DJ_k\| \le Ck^{\ell}$ in those charts. Partition the manifold into a deterministic region, where $|J_k| \ge c > 0$ on B_k, and its complement, where each pointwise distribution is Gaussian with least covariance eigenvalue at least c^2k^{-2a}. No independence across points is required. Then for any fixed $D > 0$, after coordinate norm comparisons,*

$$\mathbb{P}(B_k \cap \{\inf |J_k| \ge k^{-D}\}) \ge 1 - \mathbb{P}(B_k^c) - C_D k^{ma+d\ell-(m-d)D}. \tag{10}$$

Changing the lower target by any fixed constant changes only the constants and the threshold on k.

Proof. A Gaussian with these eigenvalue bounds has density at most Ck^{ma}, regardless of its mean. The probability of a ball of radius C_0k^{-D} is at most Ck^{ma-mD}. Use a net of mesh $c_0k^{-D-\ell}$, with $Ck^{d(D+\ell)}$ points. Choose c_0 small relative to the Lipschitz and coordinate comparison constants, and C_0 large relative to the desired lower target. At deterministic-region net points the target holds on B_k for large k. At the other net points use the *unconditional* Gaussian small-ball probabilities and a union bound. On B_k the interpolation error is at most Cc_0k^{-D}. Every point is close to a good net point, regardless of which side of the region boundary either point belongs to. Subtracting $\mathbb{P}(B_k^c)$ proves the claim. Conditioning the Gaussian law on B_k was neither done nor justified. □

Theorem 4.2 (M675 noncancellation budget)**.** *Assume the source envelope, packet, and collar contracts described above, and fix all finite orders. Then its Gaussian construction satisfies*

$$\mathbb{P}\left(B_k \cap \left\{\inf_B \frac{|u| + |\mathrm{d}u|_{g_0}}{W} \ge k^{-406}\right\}\right) \ge 1 - Ck^{-1} - Ck^{40}e^{-k^2/2} \tag{11}$$

for all sufficiently large allowed frequencies. More generally $D > 405$ gives error Ck^{405-D} in place of Ck^{-1}.

Proof. Let $w = l_+ - l_-$, $s_i = l_- + iw/5$, and $b_N(t) = N\log(1-t)/(2k)$. The source fixes the additive constant of h so that $h(s_2) = b_N(s_2)$. It proves

$$h \le b_N - cw \ (t \le s_1), \qquad h \ge b_N + cw \ (s_3 \le t \le s_4).$$

Choose $\beta = 1$ for $t \le s_3$, zero for $t \ge s_4$, and put $W = e^{kh} + \beta e^{kb_N}$. On the transition region $W/e^{kh} \le 1 + e^{-cwk}$. The packet net covers $\{t \ge s_1\}$ at mesh Ck^{-10} and has at most Ck^{40} packets. Let B_k mean that all independent real Gaussian coefficients have absolute value at most k. Then

$$\mathbb{P}(B_k^c) \le Ck^{40}e^{-k^2/2}, \qquad |D^q u| \le C_q k^{2q+41}W, \qquad |\mathrm{d}W|/W \le Ck^2. \tag{12}$$

The residual bounds are obtained by requesting order $M + 42$ from individual packets, absorbing their count and the coefficient bound.

Use the deterministic region $W/e^{kh} > k^{44}$. There $\beta = 1$ for large k. The beam value and angular derivative have combined norm at least ce^{kb_N}, while the sum of packet values and gradients on B_k is

at most $Ck^{43}e^{kh}$. Their ratio is $O(k^{-1})$, so the beam gives a lower bound cW. This region contains *all* $t < s_1$ for large k by the exponential collar gap. Consequently every point of its complement has a nearby net center. Corollary 3.2, even weakened to the source's ck^{-1}, gives least standard deviation for the unnormalized ordinary jet at least $ck^{-1}e^{kh}$. Dividing by $W \le k^{44}e^{kh}$ gives ck^{-45}. Additional independent packets add nonnegative covariance. Beam addition is only a mean shift.

For $J_k = W^{-1}(u, \partial_1 u, \dots, \partial_4 u)$, differentiating and using (12) through order two gives $|DJ_k| \le Ck^{45}$: both D^2u/W and $(\mathrm{d}W/W)(\mathrm{d}u/W)$ have this bound. There is no need to replace Ck^{45} by k^{46}. Apply Lemma 4.1 with $(d, m, a, \ell) = (4, 5, 45, 45)$. The density exponent is 225, the net count exponent is $4D + 180$, and their sum with the small-ball volume is $225 + 4D + 180 - 5D = 405 - D$. Fixed constants in the net spacing and threshold handle all metric and chart comparisons. Taking $D = 406$ proves the result. □

Theorem 4.3 (Joint-volume improvement)**.** *Under the same hypotheses as Theorem 4.2, with* $\delta = 10^{-6}$*, one has*

$$\mathbb{P}\left(B_k \cap \left\{\inf_B \frac{|u| + |\mathrm{d}u|_{g_0}}{W} \ge k^{-398}\right\}\right) \ge 1 - Ck^{-5/4+18\delta} - Ck^{40}e^{-k^2/2}. \tag{13}$$

Proof. Keep the beam threshold k^{44} and the Lipschitz exponent 45. At every point in the complementary region, the normalized selected matrix $\widehat{S}_x = W^{-1}S_x$ satisfies

$$|\det \widehat{S}_x| \ge ck^{13/4-18\delta-220} = ck^{-867/4-18\delta}.$$

The full jet covariance Σ_x dominates $\widehat{S}_x\widehat{S}_x^T$. Conjugating by the latter matrix's positive square root proves determinant monotonicity here, so $\sqrt{\det \Sigma_x} \ge |\det \widehat{S}_x|$. The maximum Gaussian density is therefore at most $Ck^{867/4+18\delta}$, not the larger Ck^{225} obtained from five copies of the weakest direction. On a net of mesh c_0k^{-D-45} the union bound is

$$Ck^{867/4+18\delta-5D+4D+180} = Ck^{1587/4+18\delta-D}.$$

At $D = 398$ this is $Ck^{-5/4+18\delta}$, which tends to zero. The deterministic-region and interpolation arguments are unchanged. This proof uses actual independent selected coefficients; a determinant of arbitrary sampled jets would not establish the covariance inequality. □

Remark 4.4 (Margins and scope)**.** A threshold $W/e^{kh} > k^{43+\eta_0}$ with fixed $\eta_0 > 0$ gives $a = 44 + \eta_0$ and the sufficient inequality $D > 400 + 5\eta_0$ using the source's weaker bank bound. Retaining $k^{-3/4}$ instead gives $a = 44 + 3/4$ for the threshold k^{44} and exponent $403.75 - D$. Theorem 4.3 uses joint volume instead. Neither calculation is claimed optimal. The integer 406 remains a useful bound using only the weaker inherited singular value. None of these exponents increases the envelope growth exponent γ or proves the envelope construction.

5 Full jets and exact coefficient correction

5.1 What the private finite-jet calculus does and does not provide

The differentiated finite-window proof in [3] starts with an exact Taylor remainder, not just a formal moment expansion. In its notation, if $Lf = \sum_{n=0}^{N} S_n f^{(n)} + \rho_N$, then

$$(Lf)^{(q)} = \sum_{p=0}^{q} \binom{q}{p} \sum_{n=0}^{N} S_n^{(q-p)} f^{(n+p)} + \rho_N^{(q)}. \tag{14}$$

When this is represented on the truncated jet $(f, \dots, f^{(N)})$, the full row remainder is

$$r_q = \rho_N^{(q)} + \sum_{p=0}^{q} \binom{q}{p} \sum_{n+p>N} S_n^{(q-p)} f^{(n+p)}.$$

The terms beyond N disappear identically for a polynomial of degree at most N, not for a general smooth function with a small value remainder. The finite jet reconstruction theorem also assumes nonsingularity of the entire jet matrix, with the full row remainder amplified by its inverse. Differentiating an integral remainder requires the stated domination of the kernel derivatives; moving integration endpoints would add boundary terms. The nonlinear jet construction evaluates the actual nonlinearity on a jet polynomial and has an explicit closure defect. A small defect is not an exact solution or a global branch.

For M675 the analogous requirements are the differentiated packet estimates (4), a lower bound for the *whole* first jet, and an exact product-rule correction. No integral-kernel representation of the manifold operator is being asserted. The next theorem performs that correction and keeps the inverse and its numerator together.

Theorem 5.1 (Homogeneous quotient correction)**.** *Fix a nonnegative integer R and $\nu \geq 0$. Let u be real and smooth and $W > 0$ on B, with residual $\mathfrak{r} = \operatorname{div}_\mu(A\mathrm{d}u) + k^2\rho u$. Assume, for $k \geq 1$,*

$$|u| + |\mathrm{d}u|_{g_0} \geq ck^{-\nu}W, \tag{15}$$

$$|D^q u| \leq C_q k^{2q+41} W \quad (0 \leq q \leq R+2), \tag{16}$$

$$|D^q \mathfrak{r}| \leq C_q k^{-M} W \quad (0 \leq q \leq R+1). \tag{17}$$

All derivatives refer to fixed finite charts or equivalent fixed covariant norms. Let $I = g_0^{-1}$, $d = u^2 + |\mathrm{d}u|_{g_0}^2$, and define

$$B_0 = \frac{\mathfrak{r}u}{d}, \quad B_1 = \frac{\mathfrak{r}I\mathrm{d}u}{d}, \quad \delta A = -B_0 I, \quad \delta\rho = k^{-2}(-B_0 + \operatorname{div}_\mu B_1). \tag{18}$$

Then u solves the exact corrected equation

$$\operatorname{div}_\mu((A + \delta A)\mathrm{d}u) + k^2(\rho + \delta\rho)u = 0, \tag{19}$$

and

$$\|\delta A\|_{C^R} + \|\delta\rho\|_{C^R} \leq C_R k^{-M+2\nu+43+(\nu+45)R}. \tag{20}$$

The corrections vanish on every open set where $\mathfrak{r} = 0$. Positivity of the corrected coefficients follows when the actual C^0 bound is smaller than their original ellipticity and density margins.

Proof. Regard $j = (u, \nabla_{g_0} u)$ as a section of $\mathbb{R} \oplus TB$, with its fixed positive fiber metric. In a fixed smooth local orthonormal frame,

$$|j| \geq c' k^{-\nu} W, \qquad |D^q j| \leq C_q k^{2q+43} W \quad (0 \leq q \leq R+1).$$

Frame derivatives are bounded fixed coefficients and are absorbed in these estimates. The map $H(z) = z/|z|^2$ is smooth away from zero and homogeneous of degree -1. By compactness of the unit sphere and homogeneity,

$$\|D^m H(z)\| \leq C_m |z|^{-m-1} \quad (m \geq 0). \tag{21}$$

For $q \geq 1$, a term in the higher chain rule for $D^q H(j)$ consists of $D^m H(j)$ applied to m derivatives of j of positive orders $q_1 + \cdots + q_m = q$, with $m \leq q$. Its bound is

$$C_q k^{(m+1)\nu+2q+43m} W^{-1} \leq C_q k^{\nu+(\nu+45)q} W^{-1}.$$

The same final bound holds for $q = 0$ directly. Multiplying by the full residual jet (17) and applying the product rule gives

$$|D^q(B_0, B_1)| \leq C_q k^{-M+\nu+(\nu+45)q}, \qquad 0 \leq q \leq R+1. \tag{22}$$

No derivative of W is taken: its powers cancel within each differentiated term. The divergence in (18) costs one derivative, followed by k^{-2}. Hence its largest exponent is $-M+\nu+(\nu+45)(R+1)-2 = -M+2\nu+43+(\nu+45)R$. This dominates the C^R exponent of δA and the nondivergence term for $\nu \geq 0$.

For exactness put $f = -\mathfrak{r}/d$. Then $\delta A = fuI$ and $k^2\delta\rho = fu - \operatorname{div}_\mu(fI\mathrm{d}u)$. The residual change is

$$\begin{aligned} \operatorname{div}_\mu(fuI\mathrm{d}u) + u(fu - \operatorname{div}_\mu(fI\mathrm{d}u)) &= f|\mathrm{d}u|^2_{g_0} + fu^2 \\ &= -\mathfrak{r}. \end{aligned}$$

This is an identity, not a small-error assertion. Since $d > 0$, all formulas are smooth. If $\mathfrak{r}$ vanishes on an open set, its product with $H(j)$ and the divergence of that product vanish there. The last assertion follows from the elementary stability of strict positive definiteness under a sufficiently small symmetric perturbation. □

Corollary 5.2 (Finite residual orders)**.** *The following sufficient choices give correction norm at most $C_R k^{-2}$:*

first-jet exponent ν	*source of lower bound*	*sufficient M*
500	*source noncancellation*	$1045 + 545R$
406	*Theorem 4.2*	$857 + 451R$
398	*Theorem 4.3*	$841 + 443R$

The source's own separate-denominator estimate at $\nu = 500$ requests $M \geq 2131 + 1088R$.

Proof. Substitute in (20) and leave an exponent of at most -2. For comparison, bounding derivatives of d by $Ck^{2q+86}W^2$ and of d^{-1} separately yields

$$|D^q d^{-1}| \leq C_q k^{2\nu+(2\nu+88)q} W^{-2}.$$

The source's product estimates then give the correction exponent $-M+4\nu+129+(2\nu+88)R$. At $\nu = 500$ this is $-M+2129+1088R$; at $\nu = 398$ it is $-M+1721+884R$. Both are valid coarser budgets. The improved bound does not differentiate the reciprocal separately from the jet numerator, which is why it avoids this duplication of inverse losses. □

Practical meaning. For a target C^R correction, request the residual through $R+1$ and the function through $R+2$, choose the sum-residual order M above, then use (5) with packet order $M+42$ and $J = R+2$. All choices precede the large-frequency limit. The constants C_R can be large; $C_R k^{-2}$ is not a finite-frequency tolerance until k is chosen accordingly. This reduces a sufficient symbolic order budget, not a measured runtime or the number of envelope generations.

6 Sign production, simplicity, and spectral persistence

6.1 Conditional variance on wavelength balls

The noncancellation bank is not by itself the sign-production argument. The source uses the additional oscillatory packet estimates at $R_x = (kQ_x)^{-1}$. A maximal disjoint family of such balls in V has bounded enlargements covering V, with Q comparable on each enlargement, since $|DQ| \le Ck^{4\delta}$ and their radii are at most k^{-1}. Therefore

$$\sum_x R_x^3 \ge c \int_V kQ(y)\,\mathrm{d}y \ge ck^{1+\gamma}. \tag{23}$$

On these balls the beam is absent. Put $S_x^2 = \sum_w |w(x)|^2$ and $U_x(Y) = u(x + R_xY)/S_x$. The nearby bank implies $S_x \ge ce^{kh(x)}$. The source scaled-packet lemma controls $\sum_w \sup_{|Y|\le 2} |D_Y^j w(x + R_xY)|^2/S_x^2$ for each required fixed j. For non-tail packets, the cutoff equals one, the amplitude is bounded away from zero, and $|\Im \mathrm{d}_Y \log w| \ge c$. Tail derivatives are exponentially small relative to $e^{kh(x)}$; they are bounded absolutely, not by dividing by a possibly zero tail value. This distinction is needed for the variance estimate.

Here is the resulting linear algebra explicitly. Let z be the row of packet values in real Gaussian coordinates and T the four-row derivative bank in Y coordinates. Then $|z| = S_x$ and

$$\mathrm{Cov}(\mathrm{d}_Y U_x \mid U_x(0)) = S_x^{-2} T(\mathrm{Id} - zz^T/S_x^2)T^T. \tag{24}$$

For a nonzero packet value, its coefficient pair has the unit quadrature vector $c_w = (-\Im w, \Re w)/|w|$. These vectors are orthonormal over distinct packets and perpendicular to z, and $Tc_w = |w|\Im(\mathrm{d}_Y w/w)$. Thus the trace in (24) is at least c after discarding exponentially small tails. One deterministic unit direction has conditional variance at least $c/4$. Upper covariance and regression coefficients are bounded by the same scaled estimates.

Gaussian regression now gives, for fixed large H, a lower bound c_1/H for $1 \le \partial_e U_x(0) \le 2$ and $|U_x(0)| < c'/H$. The derivative estimates through order five give $\mathbb{E}\,\|U_x\|_{C^2}^2 \le C$ by $H^5 \hookrightarrow C^2$ in dimension four. Subtracting $\mathbb{P}(\|U_x\|_{C^2} > H) \le C/H^2$ leaves a fixed positive probability. Taylor's theorem at centers $\pm a'e/H$, with $a' = 1/4$ and $c' < a'/4$, gives opposite values of size at least $a'/(2H)$. Disks of radius $a'/(4H^2)$ preserve the signs under the gradient bound. The centers, directions, and disks are chosen before sampling.

Let F be the fraction of the total weight $\sum R_x^3$ contributed by successful balls. Then $0 \le F \le 1$ and $\mathbb{E}F \ge \pi_* > 0$, without any independence assumption between balls. Since $\mathbb{E}F \le p + (1-p)\pi_*/2$ when $p = \mathbb{P}(F \ge \pi_*/2)$, one obtains $p \ge \pi_*/(2 - \pi_*)$. Combining with (23) gives a certificate of size $ck^{1+\gamma}$ with fixed positive probability. Subtracting the vanishing failure probability in Theorem 4.3 leaves a nonempty intersection for large k. Exact coefficient correction changes the pair, not the function or its signs.

6.2 Why the protected patch supplies simplicity

The source's packets vanish on $t < s_1 - Ck^{-1/3-100\delta}$, where $u = u_*$ and the residual is zero. Hence the correction preserves a round patch. For $N > 1$ choose a smaller product patch with $\mathrm{d}u$ and $\mathrm{d}t$ independent and a nonnegative bump χ. The tensor

$$E = \chi\left(I - \frac{(I\mathrm{d}u)\otimes(I\mathrm{d}u)}{|\mathrm{d}u|_{g_0}^2}\right)$$

is nonnegative and annihilates $\mathrm{d}u$. On the old k^2 eigenspace the quadratic form $f \mapsto \int E(\mathrm{d}f, \mathrm{d}f)\mu$ has kernel precisely $\mathbb{R}u$. Indeed zero energy implies $f = b(u)$ on the smaller patch. There

$$b''(u)|\mathrm{d}u|^2 = k^2(ub'(u) - b(u)), \qquad |\mathrm{d}u_*|^2 = N^2((1-t)^{N-1} - u^2).$$

Differentiate at fixed u with respect to t. Because $N > 1$, $b'' = 0$; substitution eliminates the constant term of b. Smooth elliptic unique continuation on the connected base then gives $f = cu$ globally. This uses unique continuation as an external elliptic input, not a finite-jet theorem.

The change $A \mapsto A + \tau E$ preserves u and k^2 exactly. If arbitrary small positive τ retained multiplicity, take other normalized eigenfunctions $f_m \perp u$ at $\tau_m \downarrow 0$. Uniform ellipticity gives an H^1 bound and a subsequence converging strongly in L^2 and weakly in H^1 to an old eigenfunction $f_0 \perp u$. Testing against every old eigenfunction cancels the original form and gives $\int E(\mathrm{d}f_m, \mathrm{d}\phi)\mu = 0$. Passing to the limit and taking $\phi = f_0$ contradicts the kernel description. Thus all sufficiently small positive τ make this eigenvalue simple; the finite C^R cost can be made arbitrarily small after k is fixed.

6.3 A weighted operator criterion for preserving the signs

The source proves existential persistence by min–max continuity and compact elliptic estimates in finite coefficient norms. The following sufficient criterion exposes the additional analytic quantities required to make that step quantitative. It is not a claim that those quantities have been numerically certified for the source's eventual metric.

Theorem 6.1 (Resolvent and sign-margin criterion)**.** *Let $\mathcal{A} = -L_{A,\rho}$ on $H = L^2(\rho\mu)$ have a simple positive eigenvalue λ, real eigenfunction $u \in H^4$, and a finite sign certificate whose minimum endpoint magnitude is $m_* > 0$. Let $g > 0$ be its distance to the rest of the spectrum, choose $0 < r < \min(g/3, \lambda/2)$, and put $\Gamma = \{|z - \lambda| = r\}$. For a second smooth positive pair define the unitary map*

$$U : L^2(\rho'\mu) \longrightarrow H, \qquad Uf = \sqrt{\rho'/\rho}\, f, \quad \mathcal{A}' = U(-L_{A',\rho'})U^{-1}.$$

Both operators have domain H^2. Set $E = \mathcal{A}' - \mathcal{A}$ and assume finite bounds

$$K_0 = \sup_\Gamma \left\|(z-\mathcal{A})^{-1}\right\|_{H\to H^2}, \qquad \delta_0 = \|E\|_{H^2\to H},$$
$$K_2 = \sup_\Gamma \left\|(z-\mathcal{A})^{-1}\right\|_{H^2\to H^4}, \qquad \delta_2 = \|E\|_{H^4\to H^2}.$$

If $\theta_0 = \delta_0 K_0 < 1/2$, $\theta_2 = \delta_2 K_2 < 1$, and

$$C_{\mathrm{emb}} \frac{K_2\delta_2}{1-\theta_2} \|u\|_{H^4} < m_*, \tag{25}$$

where $\|f\|_{C^0} \le C_{\mathrm{emb}} \|f\|_{H^4}$, the new pair has a simple eigenvalue in $(\lambda - r, \lambda + r)$ and a real eigenfunction with the same sign certificate.

Proof. Write $R(z) = (z - \mathcal{A})^{-1}$. Since ER has norm at most θ_0 on H, the resolvent of $\mathcal{A}'$ is $R'(z) = R(z)(\mathrm{Id} - ER(z))^{-1}$. The spectral distance on Γ is r, so $\|R(z)\|_{H\to H} \le 1/r$. With Riesz projections P, P', the resolvent identity and contour length give

$$\|P' - P\|_{H\to H} \le \frac{\theta_0}{1-\theta_0} < 1.$$

Two orthogonal projections at distance less than one have equal ranks: restriction of either to the other's range is injective. Here P has rank one, so P' does too. Self-adjointness puts its one eigenvalue on the real axis, and $r < \lambda/2$ makes it positive.

On H^2, the bound $\theta_2 < 1$ also gives $\|R'(z)\|_{H^2 \to H^4} \le K_2/(1-\theta_2)$, consistently with the previous inverse. Because $R(z)u = (z-\lambda)^{-1}u$,

$$\|P'u - u\|_{H^4} = \left\| \frac{1}{2\pi i} \int_\Gamma R'(z) E R(z) u \, \mathrm{d}z \right\| \le \frac{K_2 \delta_2}{1-\theta_2} \|u\|_{H^4} .$$

The real coefficients imply that $P'u$ is real. Inequality (25) preserves every strict endpoint sign and also makes $P'u$ nonzero. It spans the new one-dimensional spectral subspace. Finally $U^{-1}P'u$ is an eigenfunction of the original new weighted operator, and U^{-1} is pointwise positive, so the signs are unchanged. □

All Sobolev norms here use fixed equivalent background norms, not Euclidean coordinate norms on a finite sampled matrix. On bounded ellipticity and density sets, sufficiently small C^4 coefficient differences control δ_0, δ_2 by expanding the conjugated second-order operator and its derivatives through order two. Constants depend on the original smooth pair and density margins. The spectral gap and K_0, K_2 are full-operator data. The physical Gram discussion in [4] similarly distinguishes the correct Hilbert norm and the omitted-component residual from a finite coordinate compression. No finite positive Gram matrix supplies these infinite-dimensional resolvent bounds.

7 The fixed-metric limit and the nodal witness

We spell out the final implication to isolate what the local improvements do not prove. Assume the source one-step proposition, the simplicity input just described, and spectral persistence for every finite previous certificate. At step j choose nested round collars tending to $t = 1/4$. Let s_i be a finite coefficient regularity order sufficient for persistence at stage i, with tolerance $\varepsilon_i > 0$. At stage j request regularity

$$r_j \ge \max(j+1, s_1, \dots, s_{j-1})$$

and coefficient increment at most

$$\eta_j \le \min\left(2^{-j}, b\,2^{-j-1}, \min_{i<j} 2^{-(j-i+1)}\varepsilon_i\right),$$

where b is a margin inside the common C^1 neighborhood. All these are finite requirements before stage j is constructed. The common exponent γ is already fixed. If its certificate coefficient is $c_j > 0$, choose an allowed frequency so large that $k_j > 2k_{j-1}$ and $c_j k_j^{\gamma/2} > j$. This is where "every sufficiently large N" in the one-step proposition is used, not just one frequency for one accuracy request.

For any fixed order q, all sufficiently late increments have summable C^q norm. The pairs converge smoothly to (A_∞, ρ_∞), and their total C^1 change remains inside the neighborhood. For each fixed i, the future tail in its required norm is at most $\varepsilon_i/2$. Consequently there is a limit-pair eigenfunction at $\lambda_i \in (k_i^2/2, 2k_i^2)$ retaining the stage-i certificate of size greater than $ik_i^{1+\gamma/2}$. Distinctness and divergence follow from the frequency separation. Positivity and smoothness of the limit pair are part of this summability argument, not consequences of finite covariance.

To convert to a Laplacian, put

$$G_\infty^{-1} = A_\infty/\rho_\infty, \qquad q = \frac{\rho_\infty \mu}{\mathrm{dvol}_{G_\infty}}, \qquad g = G_\infty + q^2 \mathrm{d}\varphi^2 \quad \text{on } S^4 \times S^1. \tag{26}$$

The quotient of positive densities is a smooth positive function. Since $\mathrm{dvol}_g = \rho_\infty \mu \, \mathrm{d}\varphi$, the divergence formula gives, for functions independent of φ,

$$\Delta_g f = (\rho_\infty \mu)^{-1} \partial_a (\mu A_\infty^{ab} \partial_b f) = L_{A_\infty, \rho_\infty} f.$$

Thus the base eigenfunctions are exact eigenfunctions of this one fixed five-dimensional metric.

Proposition 7.1 (Certificate-to-measure transfer)**.** *For a fixed smooth metric (26) and the fixed relatively compact certificate chart, there is $c_g > 0$ such that every function independent of φ with certificate size S obeys $\mathcal{H}_g^4(\{u = 0\}) \geq c_g S$.*

Proof. For each point of an end disk, the segment connecting the two oppositely signed ends has a zero by continuity. In the cylinder times a fixed circle interval I, projection onto the end disk times I maps the zero set onto that product. The projection is uniformly K-Lipschitz in the fixed metric on the compact chart. The Lipschitz inequality for Hausdorff measure gives $\mathcal{H}_g^4(\{u = 0\} \cap (\text{cylinder} \times I)) \geq K^{-4} |I| \mathcal{H}^3(\text{disk})$. The cylinders lie inside disjoint balls, so these inequalities add. No regularity of the nodal set, nor an implicit-function assertion at its singular points, is required for this lower estimate. □

Apply the proposition to the retained certificates. With $\varepsilon_0 = \gamma/4$ and $\lambda_i \leq 2k_i^2$, the ratio in (2) is at least a fixed positive multiple of i, and diverges. This is a *conditional rederivation of the source's final implication.* The unresolved burden for an independent global proof remains the simultaneous validity of the full envelope induction, packet construction, all-order accuracy choices, and elliptic input across the recursive sequence. The local theorems in this report neither bypass that burden nor infer a fixed metric from metrics chosen separately for each frequency.

8 Examples and failure boundaries

A concrete bank. Take $b = 0$, $\epsilon = 1/8$, $\eta = 1/2$, and $r = \sqrt{1 + \epsilon^2}$. The real Gram block is $\left(\begin{smallmatrix} 4 & 1/8 \\ 1/8 & 1/64 \end{smallmatrix}\right)$, with determinant $3/64$. The selected minor has determinant $1/40$. Thus the bank remains full row rank even at a zero envelope gradient; it is the distinct real slope, not a division by $|p|$, that resolves the value/longitudinal direction. Removing that real-slope perturbation makes the real block rank one. Removing transverse perturbations makes its imaginary block rank one. All five rows, and the right coefficient law, are therefore substantive.

Why joint volume is better than the weakest direction. For the model covariance $\mathrm{diag}(\epsilon^2, \eta^2, \eta^2, 1, 1)$ with $0 < \epsilon \leq \eta \leq 1$, the density maximum is $(2\pi)^{-5/2}/(\epsilon\eta^2)$, whereas using five copies of the weakest standard deviation gives $(2\pi)^{-5/2}\epsilon^{-5}$. Their ratio is η^2/ϵ^4. This model illustrates the determinant mechanism, not an assertion that the actual packet covariance is diagonal.

The inverse margin cannot be omitted. For $j_a(t) = (a, t)$ with $a > 0$, $H(j_a(t)) = (a, t)/(a^2 + t^2)$. Its value at zero has norm a^{-1} and its first derivative at zero has norm a^{-2}. Higher derivatives have the corresponding homogeneity losses. Formula (21) captures this amplification. An arbitrarily small residual with no relation to a does not guarantee a small correction.

Positive covariance is not uniform nonvanishing. The scalar Gaussian field $F(x) = g + x$ on $[-1, 1]$, with g standard normal, has pointwise variance one at every point, but has a zero with positive probability. M675 escapes this example through its five real jet components on a four-dimensional base, an actual density bound, and derivative control between net points. A finite list of positive sample covariances establishes none of those missing statements between samples.

A small residual is not an exact eigenfunction. On the round circle, $f(x) = \sin(nx) + a\sin(2nx)$ has $(-\partial_x^2 - n^2)f = 3an^2\sin(2nx)$. Taking a small makes the residual small but does not make f an eigenfunction at n^2 unless $a = 0$. Theorem 5.1 deliberately changes the coefficients to make a function exact. Theorem 6.1 instead tracks a genuine nearby eigenspace under coefficient changes and needs a spectral gap. These are different operations.

Numerical order comparison. For $R = 2$, the original separate-denominator budget is $M \geq 4307$. Keeping its $\nu = 500$ lower bound but using the joint quotient gives $M \geq 2135$. Using the selected-volume bound $\nu = 398$ gives $M \geq 1727$. The corresponding per-packet residual requests are higher by 42. These integers are sufficient asymptotic exponents. They are not interval bounds on the actual correction at a specified finite frequency.

9 Public context and formalization status

Primary pages checked on October 8, 2026 include the source repository, the Annals pages for Logunov's polynomial upper estimate and lower-bound work [6, 7], the publisher page for Donnelly–Fefferman [5], Hezari's Gevrey/quasianalytic estimates [8], and the author preprint record of Dencker–Sjöstrand–Zworski [9]. The established polynomial upper estimates do not by themselves supply a sharp square-root upper bound in the smooth setting. The analytic and Gevrey/quasianalytic regularity regimes must not be identified with arbitrary smooth coefficients. The semiclassical pseudomode literature supplies context for the packet method, not the exact fixed-metric counterexample. This report does not reprove those papers or assert priority over them. Publisher metadata and abstracts were checked; proofs behind subscription previews were not claimed read.

The public repository currently describes 719 manuscripts, whereas this project preserves the initial 722 manuscript identifiers. M675 is the preserved identifier. The local manuscript is pinned to commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; the separate byte-identity audit compares it with the captured October 8 trees. Such a captured-tree comparison is not a new live-branch or mathematical verification.

Family 350's public scope page reports a Lean formalization for a fixed metric on $S^4 \times S^1$ and unbounded nodal ratio. Its linked comparator defines smooth metrics, nonzero smooth eigenfunctions with positive eigenvalues tending to infinity, and intrinsic four-dimensional Hausdorff measure; the ratio in that target is divided by $\sqrt{\lambda}$ [10, 11]. It does not state a fixed $\varepsilon_0 > 0$ in that denominator. The comparator theorem has a `sorry` placeholder, but this is a target file, not evidence that an implementation is absent. The formalization catalogue maps it to `OAI/Geometry/NodalSets/Main.lean`, whose displayed proof composes two imported smooth-limit results [12]. The comparator configuration permits `propext`, `Quot.sound`, and `Classical.choice`; the catalogue's review field says `unchecked` [13]. We inspected these public statements and the entry point, but did not build their dependency closure, inspect all axiom dependencies, or run a kernel/comparator check. Accordingly we report the public formalization assertion and its displayed scope, not an independently kernel-checked power-law theorem. The local byte ledger keeps this live status inspection separate from the preserved mathematical source.

10 Advantages, costs, and further work

Demonstrated advantages. The Gram decomposition shows precisely where real rank is gained, including the $p = 0$ case. The selected-column determinant retains the covariance volume lost by a weakest-direction-only bound and improves the noncancellation budget with the source's actual packet geometry. The homogeneous quotient then reduces the required full residual order while preserving the exact coefficient identity, protected patch, and sign certificates. The weighted resolvent criterion provides a concrete sufficient spectral persistence test with endpoint margins, and the final transfer makes the powers of k and λ explicit. These are source-specific uses of joint covariance, full differentiated remainders, and physical norms.

Costs and limitations. The smaller powers still hide data- and order-dependent constants. No numerical condition number certifies all centers or all frequencies. The Gaussian event uses independence of the actual real coefficients and no conditional density claim after truncating them. The correction can fail to preserve positivity unless its actual norm is below the ellipticity margin. Spectral persistence requires an isolated simple eigenvalue and full-operator resolvent bounds; a finite compression or a small residual is insufficient. None of the local improvements raises γ, provides a larger explicit ε_0, or independently closes the source's geometric induction. The full fixed-metric existence claim retains its inherited status.

Future extensions with explicit obstacles. An anisotropic mesh could use directional derivatives together with the full covariance ellipsoid, but would require uniform spatial control of those ellipsoids; pointwise diagonalization alone is insufficient. Non-Gaussian coefficients would require genuine small-ball density or anti-concentration input, not merely (9). A fully quantitative spectral implementation would need certified complement resolvent estimates and nonzero sign margins for each retained certificate. Better packet order bounds would require tracking the eikonal and transport constants uniformly with derivative order; the present finite-order construction supplies no such all-order optimization. Finally, independent formal verification of the power-law statement would need a target with its positive exponent and a checked dependency chain, not only the weaker displayed comparator ratio.

Contribution assessment. The demonstrated increment is C3/5: a substantial restricted improvement in source-specific first-jet probability and exact-correction budgets, with full local proofs and a quantitative persistence criterion. It is not C4 or C5 because neither a new global fixed-metric theorem nor an independently verified historical breakthrough is established. Applicability level L4 is a routing classification and is not this contribution score. Independent bounded symbolic and floating checks are recorded separately, with their precise scope; the proofs above, not samples, support the uniform local statements.

References

[1] OpenAI, *Power-law violations of Yau's nodal upper bound*, September 23, 2026. Preserved M675 source: `main.tex` and the complete `introduction`, `framework`, `envelopes`, `waves`, and `global` sections, plus bibliography. Public manuscript directory.

[2] M. Subbarao, *Joint-Moment Bayesian Neural Inference and Inverse Uncertainty Propagation*, TR-core-pat-16, confidential author-supplied manuscript. Covariance positivity, Gaussian prop-

agation, same-law remainder, and rank-loss proofs; exact local read ranges in the evidence ledger.

[3] M. Subbarao, *Ram nonlinear integral-equation/PDE report*, `TR1-lin-nonlin-rev2/sec-03-jets.tex`, confidential author-supplied manuscript. Taylor integral remainder, full Leibniz rows, differentiated remainder, finite-jet recovery, and nonlinear closure proofs.

[4] M. Subbarao, *Ram Calculus*, Volume 21, `ch01d-physical-metrics.tex`, confidential author-supplied manuscript. Physical Gram metrics and full versus compressed residuals; complete chapter read.

[5] H. Donnelly and C. Fefferman, Nodal sets of eigenfunctions on Riemannian manifolds, *Inventiones Mathematicae* 93 (1988), 161–183. https://doi.org/10.1007/BF01393691.

[6] A. Logunov, Nodal sets of Laplace eigenfunctions: polynomial upper estimates of the Hausdorff measure, *Annals of Mathematics* 187 (2018), 221–239. https://annals.math.princeton.edu/2018/187-1/p04.

[7] A. Logunov, Nodal sets of Laplace eigenfunctions: proof of Nadirashvili's conjecture and of the lower bound in Yau's conjecture, *Annals of Mathematics* 187 (2018), 241–262. https://annals.math.princeton.edu/2018/187-1/p05.

[8] H. Hezari, Upper bounds on the size of nodal sets for Gevrey and quasianalytic Riemannian manifolds, *Communications in Mathematical Physics* 404 (2023), 1341–1359. https://doi.org/10.1007/s00220-023-04853-z.

[9] N. Dencker, J. Sjöstrand, and M. Zworski, Pseudospectra of semi-classical (pseudo)differential operators, *Communications on Pure and Applied Mathematics* 57 (2004), 384–415. https://arxiv.org/abs/math/0301242.

[10] OpenAI, *Yau's nodal bounds: surfaces and higher dimensions*, public formalization scope, inspected October 8, 2026. Family 350 scope page.

[11] OpenAI, *YauCounterexample comparator target and configuration*, inspected October 8, 2026. Target definitions and statement; Comparator configuration.

[12] OpenAI, *NodalSets/Main.lean*, public implementation entry point, inspected October 8, 2026. Implementation entry point.

[13] OpenAI, public mathematical repository and formalization catalogue, inspected October 8, 2026. Repository status; Declaration mapping and review field.